Compiler Construction Lab - Assignment 12

Name: Dev Vachhani
PRN.: 24070122517
Batch: B3

Objective

To write a program using Flex and Bison to generate intermediate code for an if-else control structure.

Theory

Intermediate code generation is an important phase of compiler design in which source code is converted into an intermediate representation before machine code is generated.

Control structures such as if-else, for, and while determine the flow of program execution. To represent these structures in intermediate code, compilers commonly use conditional jumps, unconditional jumps, and labels.

An if-else statement evaluates a condition and executes one block if the condition is true and another block if it is false. During intermediate code generation, the condition is translated into a conditional jump. If the condition is false, control transfers to the else block. After executing the if block, an unconditional jump skips the else block.

For example:

if (a < b)

x = 10;

else x =

20;

The corresponding intermediate code is:

if a < b goto L1

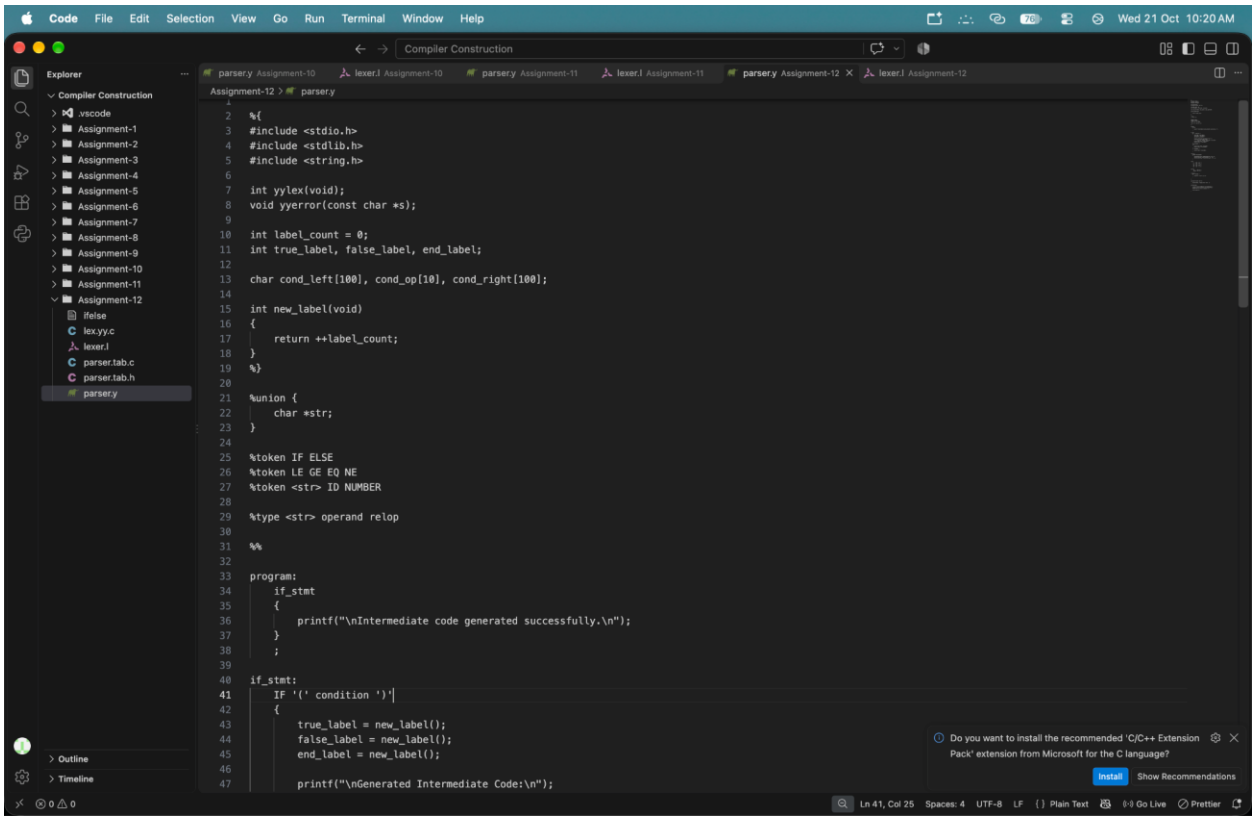
goto L2 L1: x = 10

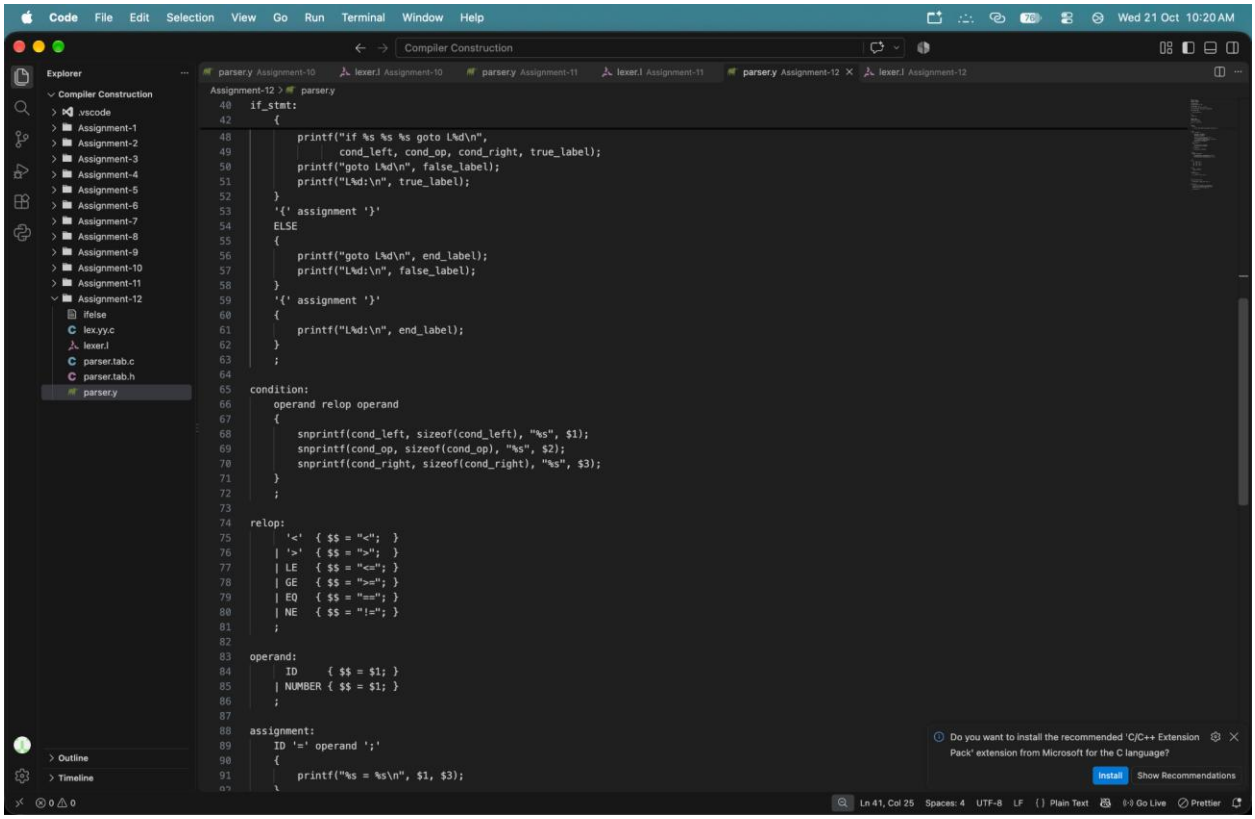
goto L3

L2: x = 20 L3:

Here, L1, L2, and L3 are labels representing different points in the program. Flex identifies tokens such as keywords, identifiers, constants, and operators, while Bison parses the input according to grammar rules and generates intermediate code through semantic actions.

Screenshots and Execution Results





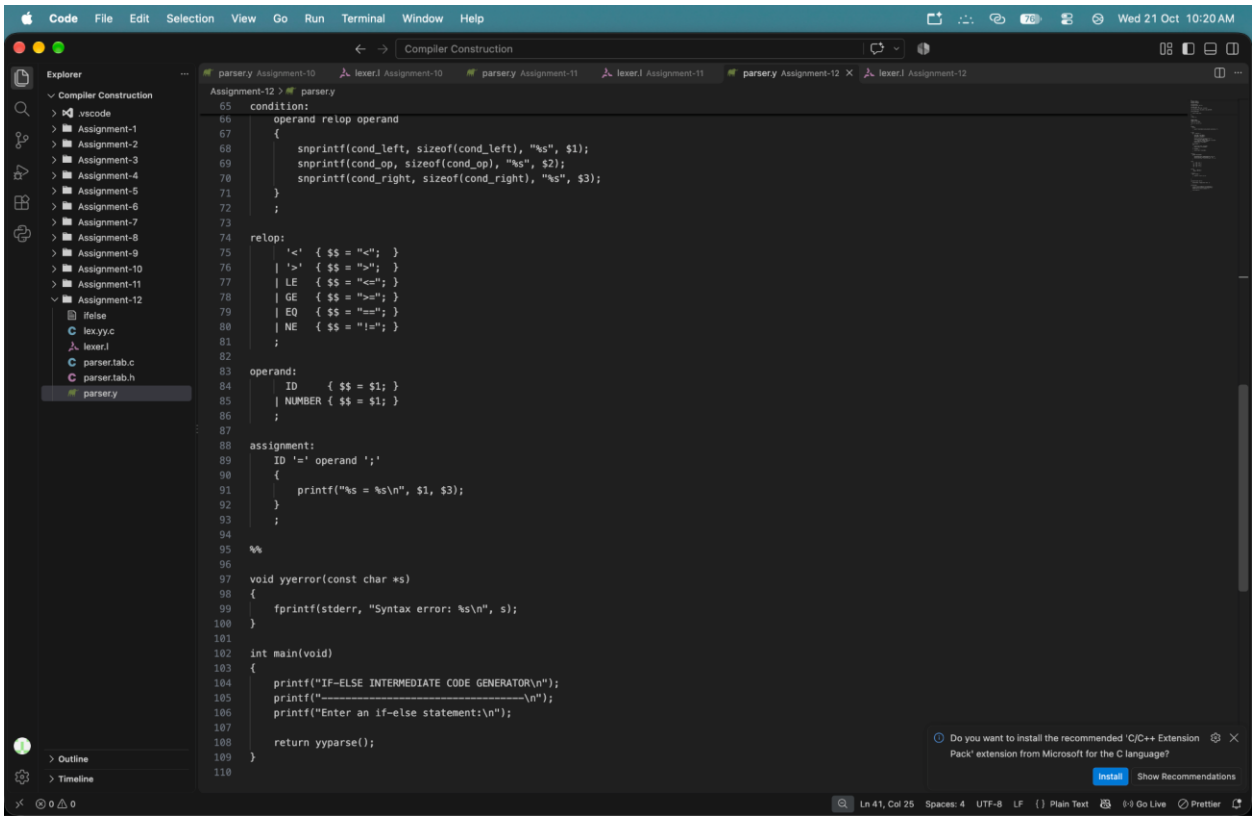
```
if_stmt:
    {
        printf("if %s %s %s goto L%d\n",
               cond_left, cond_op, cond_right, true_label);
        printf("goto L%d\n", false_label);
        printf("L%d:\n", true_label);
    }
    '{' assignment '}'
    ELSE
    {
        printf("goto L%d\n", end_label);
        printf("L%d:\n", false_label);
    }
    '{' assignment '}'
    {
        printf("L%d:\n", end_label);
    }
    ;

condition:
    operand relop operand
    {
        snprintf(cond_left, sizeof(cond_left), "%s", $1);
        snprintf(cond_op, sizeof(cond_op), "%s", $2);
        snprintf(cond_right, sizeof(cond_right), "%s", $3);
    }
    ;

relop:
      '<'  { $$ = "<";  }
    | '>'  { $$ = ">";  }
    | LE   { $$ = "<="; }
    | GE   { $$ = ">="; }
    | EQ   { $$ = "=="; }
    | NE   { $$ = "!="; }
    ;

operand:
      ID     { $$ = $1; }
    | NUMBER { $$ = $1; }
    ;

assignment:
    ID '=' operand ';'
    {
        printf("%s = %s\n", $1, $3);
```



```
condition:
    operand relop operand
    {
        snprintf(cond_left, sizeof(cond_left), "%s", $1);
        snprintf(cond_op, sizeof(cond_op), "%s", $2);
        snprintf(cond_right, sizeof(cond_right), "%s", $3);
    }
    ;

relop:
      '<'  { $$ = "<";  }
    | '>'  { $$ = ">";  }
    | LE   { $$ = "<="; }
    | GE   { $$ = ">="; }
    | EQ   { $$ = "=="; }
    | NE   { $$ = "!="; }
    ;

operand:
      ID     { $$ = $1; }
    | NUMBER { $$ = $1; }
    ;

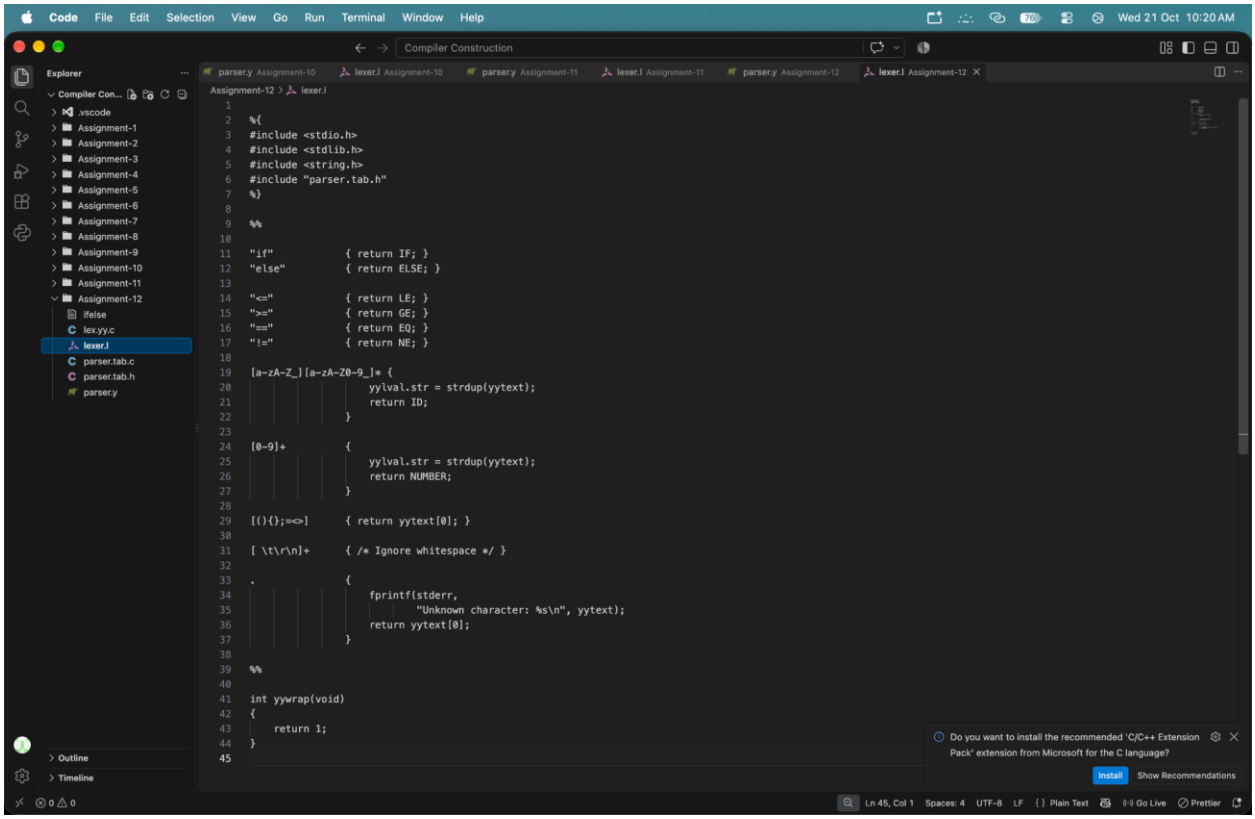
assignment:
    ID '=' operand ';'
    {
        printf("%s = %s\n", $1, $3);
    }
    ;

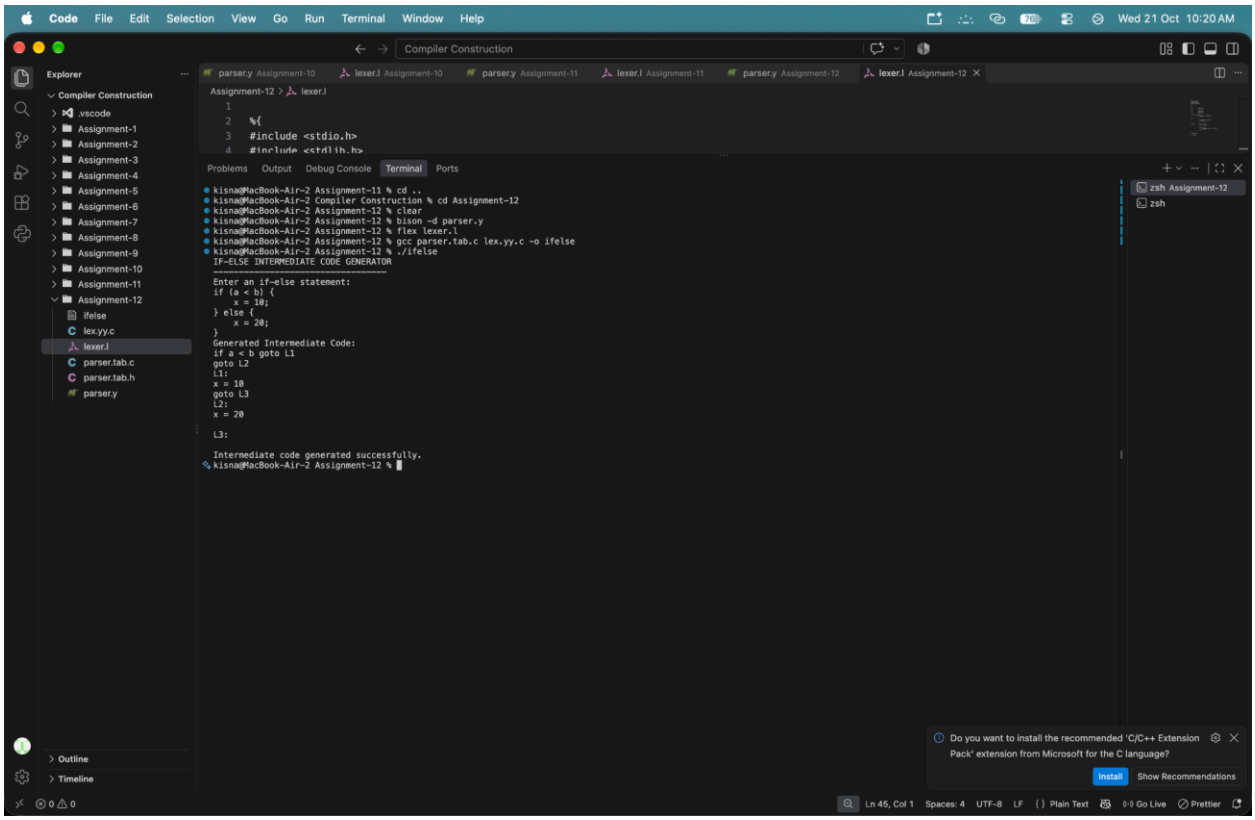
%%

void yyerror(const char *s)
{
    fprintf(stderr, "Syntax error: %s\n", s);
}

int main(void)
{
    printf("IF-ELSE INTERMEDIATE CODE GENERATOR\n");
    printf("-----------------------------------\n");
    printf("Enter an if-else statement:\n");

    return yyparse();
}
```


Code File Edit Selection View Go Run Terminal Window Help
Wed 21 Oct 10:20 AM
Compiler Construction
Explorer
Compiler Con...
.vscode
Assignment-1
Assignment-2
Assignment-3
Assignment-4
Assignment-5
Assignment-6
Assignment-7
Assignment-8
Assignment-9
Assignment-10
Assignment-11
Assignment-12
ifelse
lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
parser.y Assignment-10
lexer.l Assignment-10
parser.y Assignment-11
lexer.l Assignment-11
parser.y Assignment-12
lexer.l Assignment-12
Assignment-12 > lexer.l
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "parser.tab.h"
%}
%%
"if" { return IF; }
"else" { return ELSE; }
"<=" { return LE; }
">=" { return GE; }
"==" { return EQ; }
"!=" { return NE; }
[a-zA-Z_][a-zA-Z0-9_]* {
yylval.str = strdup(yytext);
return ID;
}
[0-9]+ {
yylval.str = strdup(yytext);
return NUMBER;
}
[(){};=<>] { return yytext[0]; }
[\t\r\n]+ { /* Ignore whitespace */ }
. {
fprintf(stderr,
"Unknown character: %s\n", yytext);
return yytext[0];
}
%%
int yywrap(void)
{
return 1;
}
Outline
Timeline
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
Install
Show Recommendations
Ln 45, Col 1
Spaces: 4
UTF-8
LF
Plain Text
Go Live
Prettier


Code File Edit Selection View Go Run Terminal Window Help
Wed 21 Oct 10:20 AM
Compiler Construction
Explorer
Compiler Construction
.vscode
Assignment-1
Assignment-2
Assignment-3
Assignment-4
Assignment-5
Assignment-6
Assignment-7
Assignment-8
Assignment-9
Assignment-10
Assignment-11
Assignment-12
ifelse
lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
Assignment-12 > lexer.l
%{
#include <stdio.h>
Problems
Output
Debug Console
Terminal
Ports
kisna@MacBook-Air-2 Assignment-11 % cd ..
kisna@MacBook-Air-2 Compiler Construction % cd Assignment-12
kisna@MacBook-Air-2 Assignment-12 % clear
kisna@MacBook-Air-2 Assignment-12 % bison -d parser.y
kisna@MacBook-Air-2 Assignment-12 % flex lexer.l
kisna@MacBook-Air-2 Assignment-12 % gcc parser.tab.c lex.yy.c -o ifelse
kisna@MacBook-Air-2 Assignment-12 % ./ifelse
IF-ELSE INTERMEDIATE CODE GENERATOR
Enter an if-else statement:
if (a < b) {
x = 10;
} else {
x = 20;
}
Generated Intermediate Code:
if a < b goto L1
goto L2
L1:
x = 10
goto L3
L2:
x = 20
L3:
Intermediate code generated successfully.
kisna@MacBook-Air-2 Assignment-12 %
zsh Assignment-12
zsh
Outline
Timeline
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
Install
Show Recommendations
Ln 45, Col 1
Spaces: 4
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